Divisibility

a divides b, $a \neq 0$.

$\Rightarrow a/b \Rightarrow b = ka$, for some k

eg: $45 = 4 \times (11) + (1) \rightarrow$ remainder

quotient

$$4 \overline{)45} \quad 11, \text{ remainder } 1$$

$$\boxed{b = ka + r}$$

Fundamental Theorem of Arithmetic

Every integer greater than 1 can be written uniquely as a prime or as a pdt of two or more primes, where the prime factors are written in order of non-decreasing size.

Q: Write the prime factorization of following numbers:

(1) 100

$2 \overline{)100}$, $5 \overline{)50}$, $2 \overline{)10}$, 5

$\Rightarrow 2^2 . 5^2$

(2) 641

$1 \overline{)641}$, 641

$\Rightarrow 641^1$

(3) 999

$3 \overline{)999}$, $3 \overline{)333}$, $3 \overline{)111}$, 37

$3^3 . 37^1$

(4) 1024

$2 \overline{)1024}$, $2 \overline{)512}$, $\ldots$

$= 2^{10}$

Divisors of a given Number

If, $N = p_1^{n_1} p_2^{n_2} \cdots p_s^{n_s}$

- The no. of divisors of N is $(n_1+1)(n_2+1)\cdots(n_s+1)$

eg: $100 = 2^2 \cdot 5^2$

no. of divisors $= (2+1)(2+1) = (3)(3) = 9$

$= 9$

1, 2,

Q: Find the no. of divisors of 4116

2	4116
2	2058
3	1029
7	343
7	49
	7

$\Rightarrow 2^2 \cdot 3^1 \cdot 7^3$

$= (2+1)(1+1)(3+1)$

$= 24$

Greatest Common Divisors (GCD)

Let 'a' & 'b' be the integers not both zero, then the largest integer 'd' such that $d|a$ & $d|b$ is called the greatest common divisor of a & b.

$\gcd(a,b) = d$

- The integers 'a' and 'b' are set to be relatively prime, if $\gcd(a,b) = 1$

- The integers $a_1, a_2, \ldots, a_n$ are pairwise relatively prime $\gcd(a_i, a_j) = 1$, $1 \le i < j \le n$

Method 1 (By Enumerating all positive divisors)

↳ Find gcd (24, 36)

24 : 1, 2, 3, 4, 6, 8, (12), 24

36 : 1, 2, 3, 4, 6, 9, (12), 18, 36

Method 2 (Using the prime factorization of integers)

Suppose that the prime factorizations of the +ve integers a and b are

$$a = p_1^{a_1} p_2^{a_2} \cdots p_s^{a_s}$$

$$b = p_1^{b_1} p_2^{b_2} \cdots p_s^{b_s}$$

where each exponent is a non-ve integer & were all primes occuring the prime factorizations of either a or b are included both factorizations with zero exponents if necessary. Then gcd (a, b) is given by

$$\gcd(a,b) = p_1^{\min\{a_1,b_1\}} \cdot p_2^{\min\{a_2,b_2\}} \cdots p_s^{\min\{a_s,b_s\}}$$

eg: gcd (24, 36)

24	
2	24
2	12
2	6
3	3
	1

$= 2^3 \cdot 3^1$

36	
2	36
2	18
3	9
3	3
	1

$2^2 \cdot 3^2$

$$\gcd = 2^{\min\{3,2\}} \cdot 3^{\min\{1,2\}} = 2^2 \cdot 3^1 = 12$$

Assignment Q: write an algorithm to find the greatest common devisors of 2 +ve integers and explain its complexity.

Step 1 : Take two +ve integers a and b

Step 2 : int a, b

Step 3 : to find out the sum, s = 0

Step 4 : then to calculate the gcd, int gcd

step 5: To find out the prime factorization of a

$$a = p_1^{a_1} \cdot p_2^{a_2} \cdots p_s^{a_s}$$

step 6: To find out the prime factorization of b

$$b = p_1^{b_1} \cdot p_2^{b_2} \cdots p_s^{b_s}$$

step 7: To find gcd.

$$\gcd(a,b) = p_1^{\min\{a_1, b_1\}} \cdots p_s^{\min\{a_s, b_s\}}$$

Q: Find gcd (252, 198)

Always take the bigno. and divide with the small no.

$252 = 1 \cdot 198 + 54$

$198 = 3 \cdot 54 + 36$

$54 = 1 \cdot 36 + \boxed{18}$ → gcd

$36 = 2 \cdot 18 + 0$

last non-zero devisor is the gcd.

$\gcd(252, 198) = 18$

This is method is Euclidean algorithm.

Q: Find the gcd of (408, 1032)

$1032 = 2 \cdot 408 + 216$

$408 = 1 \cdot 216 + 192$

$216 = 1 \cdot 192 + \boxed{24}$

$192 = 8 \cdot 24 + 0$

$\gcd(408, 1032) = 24$

Q: Show that 2201 and 120 are relatively prime using eclidean algorithm.

$2201 = 18 \cdot 120 + 41$

$120 = 2 \cdot 41 + 38$

$41 = 1 \cdot 38 + 3$

$38 = 12 \cdot 3 + 2$

$3 = 1 \cdot 2 + \boxed{1}$

$2 = 2 \cdot 1 + 0$

The gcd $(2201, 120) = 1$

gcds as linear Combinations

If 'a' and 'b' are +ve integers ($a, b \in \mathbb{Z}^+$) then their gcd exists integers 's' and 't' (s.t $\in \mathbb{Z}$) such that

$\gcd(a,b) = s \cdot a + t \cdot b$ (By Extended Euclidean)

$a, b \in \mathbb{Z}^+$, $s, t \in \mathbb{Z}$

Q: Express the gcd $(2201, 120)$ as a linear combination of 2201 & 120.

Steps to find gcd.

We have gcd $(2201, 120) = 1$

Reverse the steps of the gcd to write the above form

$\Rightarrow 1 = 3 - 1 \cdot 2$

$= 3 - 1 \cdot [38 - 12 \cdot 3]$ $= 1 \times 3 - 1 \times 38 + 12 \times 3$ $= 13 \times 3 - 1 \times 38$

$= 13 \cdot 3 - 1 \cdot 38$

$= -1 \times 38 + 13[41 - 1 \times 38] = -1 \times 38 + 13 \times 41 - 13 \times 38$

$= 13 \times 41 - 14 \times 38$

$= 13 \cdot 41 - 14 \cdot 38$

$= 13 \cdot 41 - 14[120 - 2 \cdot 41] = 13 \cdot 41 - 14 \times 120 + 28 \cdot 41$

$= 41 \cdot 41 - 14 \cdot 120$

$= 41[2201 - 18 \cdot 120] - 14 \cdot 120$

$= 41 \cdot 2201 - 752 \cdot 120$

$\gcd(2201, 120) = (41)(2201) + (-752)(120)$

Q: Find gcd (3056, 2340) using Euclideans algorithm and find 2 integers x and y such that gcd (3056, 2340) = 3056x + 2340y.

$3056 = 1 \cdot 2340 + 716$

$2340 = 3 \cdot 716 + 192$

$716 = 3 \cdot 192 + 140$

$192 = 1 \cdot 140 + 52$

$140 = 2 \cdot 52 + 36$

$52 = 1 \cdot 36 + 16$

$36 = 2 \cdot 16 + \boxed{4}$

$16 = 4 \cdot 4 + 0$

$\gcd(3056, 2340) = 4$

By extended Euclideans:

$4 = 36 - 2 \cdot 16$

$= 36 - 2 \cdot [52 - 1 \cdot 36] = 1 \times 36 - 2 \times 52 + 2 \times 36$

$= 2 \cdot 36 - 2 \cdot 52$

$= -2 \cdot 52 + 3 \cdot 36$

$= -2 \cdot 52 + 3 \cdot [140 - 2 \cdot 52] = -2 \cdot 52 + 3 \cdot 140 - 6 \cdot 52$

$= 3 \cdot 140 - 8 \cdot 52$

$= -8 \cdot 52 + 2 \cdot 140 - 6 \cdot [192 - 1 \cdot 140] = -2 \cdot 56 + 2 \cdot 140 - 4 \cdot 192 + 4$

$= -2 \cdot 56 + 6 \cdot 140 - 4 \cdot 192$

$= -2 \cdot 56 + 6 \cdot 140 - 4 \cdot [2340 - 3 \cdot 716] = -2 \cdot 56 + 6 \cdot 140 - 4 \cdot 2340 + 12 \cdot 716$

$= 2 \cdot 56 + 6 \cdot 140 - 4 \cdot 2340 + 12 \cdot [3056 - 1 \cdot 2340]$

$= 2 \cdot 56 + 6 \cdot 140 - 4 \cdot 2340 + 12 \cdot 3056 - 12 \cdot 2340$

$= 3 \cdot 140 - 8 \cdot [192 - 1 \cdot 140] = 3 \cdot 140 - 8 \cdot 192 + 8 \cdot 140$

$= 11 \cdot 140 - 8 \cdot 192$

$= -8 \cdot 192 + 11 \cdot [716 - 3 \cdot 192] = -8 \cdot 192 + 11 \cdot 716 - 33 \cdot 192$

$= -41 \cdot 192 + 11 \cdot 716$

$= 11 \cdot 716 - 41 \cdot [2340 - 3 \cdot 716] = 11 \cdot 716 - 41 \cdot 2340 + 123 \cdot 716$

$= 134 \cdot 716 - 41 \cdot 2340$

$= 134 \cdot [3056 - 1 \cdot 2340] - 41 \cdot 2340 = 134 \cdot 3056 - 134 \cdot 2340 - 41 \cdot 2340$

$= 134 \cdot 3056 - 175 \cdot 2340$

$4 = (134)(3056) + (-175)(2340)$

(∵ we multiplied with signs)

$\Rightarrow 134(8056) - 175(8310)$

$x = 184$ & $y = 175$

6/8/2026

Congruences

'a divide by n gives b'

$a \equiv b \pmod{n}$

Let n be a fixed positive integer

Two integers a, b are said to be congruent modulo n, symbolized by $a \equiv b \pmod{n}$ if n divides the difference $a - b$; that is, $a - b = kn$ for some integer k

$21 \equiv ? \pmod 5$

$21 - ? = M(5)$

$21 - 1 = 20 = M(5)$

$21 \leq (-4) \geq 25 = M(5)$

$21 \equiv -4 \pmod 5$

$21 \equiv 1 \pmod 5$

$-4 \equiv 1 \pmod 5$

$-4 - 1 = M(5)$

Ex: Given an integer a, let q and r be its quotient and remainder upon division by n, so that

$$a = qn + r, \quad 0 \leq r < n.$$

Then, by definition of congruence, $a \equiv r \pmod n$.

Because there are n choices for r, we see that every integer is congruent modulo n to exactly one of the values $0, 1, 2, \ldots, n-1$

The set of n integers $0, 1, 2, \ldots, n-1$ is called the set of least non negative residues modulo n.

$[0]$ = residue class 0 = set of all

Theorem:

∃ a arbitary integer a and b, $a \equiv b \pmod n$ if and only if a and b leave the same negative remainder when divided by n.

$a \equiv b \pmod n \Rightarrow a-b = M(n) \Rightarrow a-b = kn \Rightarrow a = b+$

Suppose, remainder

$b = q_1 n + r$, $0 \le r < n$

$\Rightarrow a = b + kn = q_1 n + r + kn = r + (q_1 + k)n$

$a = (q_1 + k)n + r$ (remainder)

$a = q_1 n + r$, $b = q_2 n + r$, $0 \le r < n$ (same rem

$a - b = M(n)$

$a - b = [q_1 n + r] - [q_2 n + r] = (q_1 - q_2)n + r - r$

$\Rightarrow a - b = (q_1 - q_2)n \Rightarrow n \mid a-b \Rightarrow a \equiv b \pmod n$

Assignment

Theorem:

Let $n > 1$ be fixed and a, b, c, d be arbitary integers. Then the following properties hold:

(a) $a \equiv a \pmod n$

(b) If $a \equiv b \pmod n$ then $b \equiv a \pmod n$

$a \equiv b \pmod n \Rightarrow a +$

$b \equiv a \pmod n$

$b - a = k(a-b) =$

$\Rightarrow n \mid b-a \Rightarrow$

(c) If $a \equiv b \pmod n$ and $b \equiv c \pmod n$, then $a \equiv c \pmod n$

(d) If $a \equiv b \pmod n$ and $c \equiv d \pmod n$, then $a+c \equiv b+d \pmod n$ and $ac \equiv bd \pmod n$

(e) If $a \equiv b \pmod n$, then $a+c \equiv b+c \pmod n$ and $ac \equiv bc \pmod n$.

(f) If $a \equiv b \pmod n$, then $a^k \equiv b^k \pmod n$ for any positive integer k.

Q: Find the remainder when 2^{20} is divided by 41

$2^{20} = 1 \pmod{41}$

$2^5 \equiv 32 \equiv (-9) \pmod{41}$

$2^5 \equiv (-9) \pmod{41}$

nearest to 41 from both sides

$2^5 = 32$ nearest of 41 common

$32 - 41 = -9$ mod 41

$2^{20} = (2^5)^4 \equiv (-9)^4 \pmod{41} = 81 \times 81 \pmod{41}$

$\equiv (-1)(-1) \pmod{41}$

$= 1 \pmod{41}$

$81 = 1 \pmod{41}$

$81 - (-1) = 82 = M(41)$

another ways Q >

$2^{20} - 1 = M(41)$

Q: Find the remainder obtained upon dividing the sum $1! + 2! + 3! + 4! + \cdots + 99! + 100!$ by 12.

Note that $4! = 24 \equiv 0 \pmod{12}$.

$4! = 1 \times 2 \times (3 \times 4)$ → factors)

$4! = 0 \pmod{12}$

Thus, for $k \geq 4$, $k! = 4! \cdot 5 \cdot 6 \cdots k \equiv 0 \pmod{12}$

There,

$1! + 2! + 3! + 4! + \cdots + 99! + 100! \equiv 1! + 2! + 3! + 0$ (1! + 1×2 + 1×2×3 + 0)

$\equiv 9 \pmod{12}$

Linear Congruence

A congruence of the form,

$$ax \equiv b \pmod{m},$$

where m is a positive integer, a and b are integers, x is a variable, is called a linear congruence.

Theorem:

If a and m are relatively prime integers and $m > 1$, then an inverse of a modulo m exists. Furthermore, the inverse is unique modulo m.

To find an inverse of a modulo m, when $\gcd(a, m) = 1$, we find a linear combination $sa + tm = 1$ where s, t are integers. Then s is an inverse of a modulo m.

Q: Find an inverse of 101 modulo 4620.

gcd(101, 4620)

$4620 = 45 \cdot 101 + 95$

$101 = 1 \cdot 75 + 26$

$75 = 2 \cdot 26 + 23$

$26 = 1 \cdot 23 + 3$

$23 = 7 \cdot 3 + 2$

$3 = 1 \cdot 2 + 1$

$2 = 2 \cdot 1$

gcd = 1.

gcd

$1 = 3 - 1 \cdot 2$

$= 3 - 1 \cdot (23 - 7 \cdot 3) = -1 \cdot 23 + 8 \cdot 3$

$= -1 \cdot 23 + 8 \cdot (26 - 1 \cdot 23) = 8 \cdot 26 - 9 \cdot 23$

$= 8 \cdot 26 - 9 \cdot (75 - 2 \cdot 26) = -9 \cdot 75 + 26 \cdot 26$

$= -9 \cdot 75 + 26 \cdot (101 - 1 \cdot 75) = 26 \cdot 101 - 35 \cdot 75$

$= 26 \cdot 101 - 35 \cdot (4620 - 45 \cdot 101) = -35 \cdot 4620 + 1601 \cdot 101$

$1 = (-35)(4620) + (1601)(101)$

Thus, 1601 is an inverse of 101 modulo 4620.

H.W

Q: Find the remainders when 2^{50} and 41^{65} are divided by 7.

$2^{50} \equiv __ \pmod 7$

$2^3 \equiv 8 \equiv 1 \pmod 7$

$(2^3)^{16} 2^2 \equiv (1)^{16} \cdot 2^2 \pmod 7$

$\equiv 1 \times 4 \pmod 7$

$\equiv 4 \pmod 7$

$\equiv 4$

$41^{65} \equiv __ \pmod 7$

$41 \equiv 6 \pmod 7$

$6 \equiv -1 \pmod 7$

$41^{65} \equiv -1 \pmod 7$

$41^{65} \equiv (-1)^{65} \pmod 7$

$41^{65} \equiv -1 \pmod 7$ $\Rightarrow -1 + 7 = 6.$

$-1 \equiv 6 \pmod 7$

$6 - (1) = \text{mod } 7$

$41^{65} \equiv 6 \pmod 7.$

$\Rightarrow 6$

Q: What is the remainder when the sum $1^5 + 2^5 + 3^5 + \cdots + 99^5 + 100^5$ is divided by 4?

$1^5 = 1 \times 1 \times 1 \times 1 \times 1 = 1$

$1^5 \equiv 1 \pmod 4$

$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

$2^5 \equiv 0 \pmod 4$

$3^5 = 3 \times 3 \times 3 \times 3 \times 3 = 243$

$\equiv 3 \pmod 4$

$4^5 \equiv 0 \pmod 4$

$1 + 0 + 3 + 0 \equiv 4 \pmod 4$

$\equiv 0 \pmod 4$

$4 \overline{)32}$ (8), 32, 0, 2

Q: prove the integer $53^{103} + 103^{53}$ is divisible by 39, and then $111^{333} + 333^{111}$ is divisible by 7.

$3 \times 13 = 39$

so should should both 3 and 13

$53^{103} \equiv ? \pmod{39}$ ← remainder

$53 \equiv 14 \pmod{39}$

$53^2 \equiv 14^2 \pmod{39}$

$53^2 \equiv 1 \pmod{39}$

$(53^2)^{51} \equiv 1^{51} \pmod{39}$

$39 \overline{)53}$ (14) ← remainder

$2 \overline{)103}$ (51), 10, 3, 2, 1

$39 \overline{)176}$ (5), 195, (1)

$(53)^{103} \equiv 1 \pmod{29}$

$= 53^{103} \cdot 53 = 53^{103} \equiv 6^{3} \pmod{29} \equiv 1 \pmod{29}$

103^{38} remainder

$\equiv 25 \pmod{29}$

$(53^{103} + 103^{38}) \bmod 29 \equiv (1+25) \bmod 29 \equiv 0 \pmod{29}$

$111^{333} + 333^{111} = M(7)$

$111 \equiv 6 \pmod{7} \equiv -1 \pmod{7}$

$\Rightarrow 111^{333} \equiv (-1)^{333} \pmod{7} \equiv -1 \pmod{7}$

$333 \equiv 4 \pmod{7}$

$\Rightarrow 333^{3} \equiv 4^{3} \pmod{7} \equiv 64 \pmod{7} \equiv 1 \pmod{7}$

$= (333^{3})^{37} \equiv 1^{37} \pmod{7}$

$\therefore 111^{333} + 333^{111} \equiv (-1+1) \pmod{7} \equiv 0 \pmod{7}$

4/08/2026

Solving Congruence

Theorem: $ax \equiv b \pmod{m}$ has a solution if and only if $d \mid b$ (d divides b), where $d = \gcd(a, m)$.

If 'd' divides 'b', then it has d mutually incongruent solutions modulo 'm'. If x_0 is any solution of $ax \equiv b \pmod{m}$ then the d mutually incongruent solutions are given by

$$x_0, x_0 + \frac{m}{d}, x_0 + 2\left(\frac{m}{d}\right), x_0 + 3\left(\frac{m}{d}\right), \ldots x_0 + (d-1)\left(\frac{m}{d}\right)$$

Q: If $d = \gcd(a, m) = 1$, then the linear congruence has a unique solution modulo m. Once we have an inverse $\bar{a}$ of a modulo m, we can solve the congruence by multiplying both sides of the linear congruence by $\bar{a}$.

$ax \equiv b \pmod m$

multiply both sides with $\bar{a}$

$\bar{a}(ax) \equiv \bar{a}b \pmod m$

$1x \equiv \bar{a}b \pmod m$

$\Rightarrow x \equiv \bar{a}b \pmod m$

$\Rightarrow \boxed{x = M(m) + \bar{a}b}$

linear congruence

$\gcd(a, m) = 1$

$\bar{a}$ is the inverse of a.

Q: what are the solutions of the linear congruence

$3x \equiv 4 \pmod 7$

$3x \equiv 4 \pmod 7$ ——①

$a = 3, b = 4, m = 7$

$d = \gcd(a, m) \Rightarrow \gcd(3, 7) = 1$

$d = 1$ and d divides b

given linear congruence has a unique

To find a solution, first find an inverse $\bar{a}$ of $a = 3$ modulo 7 by expressing 1 as linear combination of 3 and 7

By Euclidean algorithm,

$\gcd(3, 7)$

$7 = 2 \cdot 3 + 1$

$3 = 3 \cdot 1 + 0$

$\gcd(3, 7) = 1$

By extended euclidean

$1 = 7 - (2)(3)$

$1 = (1)(7) + (-2)(3)$ {How $\Rightarrow (-2)(3) + (1)(7)$}

∴ the inverse of 3 is -2.

$\bar{a} = -2$ is the inverse of 3 modulo 7

Now multiply both sides by -2

① $\times -2$ $\Rightarrow (-2)(3x) \equiv (-2)(4) \pmod 7$ ← replace existing terms

$\Rightarrow -6x \equiv -8 \pmod 7$, [$-6 \equiv 1 \pmod 7$, $8 \equiv 1 \pmod 7$]

$\Rightarrow x \equiv -8 \pmod 7$

$x \equiv -8 \pmod 7$

$\Rightarrow x + 8 = M(7)$

$\Rightarrow \boxed{x = M(7) - 8}$

$x \equiv -8 \pmod 7$

$-8 \equiv -1 \pmod 7$

$-8+1 = -7 \bmod 7$

$= 0$

The solutions to the congruence are the integers 'x' such that, $x \equiv -8 \pmod 7$ or $x \equiv 6 \pmod 7$ or $x \equiv -1 \bmod 7$ namely $6, 13, 20 \cdots \cdots -1, -8$

Q: Solve $18x \equiv 30 \pmod{42}$

$a = 18, \; b = 30, \; M = 42$

$\gcd(a, m) = \gcd(18, 42)$

By ~~ele~~ euclidean algorithm

$\gcd(18, 42)$

$42 = 2 \cdot 18 + 6$

$18 = 3 \cdot 6 + 0$

$\gcd(18, 42) = 6 \quad \therefore$

~~By extended euclidian algthm~~

~~$6 = 42 - 2 \cdot 18$~~

$18 \overline{)42} \; 2$, 36, 6

$6 \mid 42 \Rightarrow$ given congruence has 6 mutually incongruent solutions modulo 42

$18x - 30 = M(42)$

Put values to x (dbt) why we take $x = 4$ and not 1.

$x = 4 \Rightarrow 18 \times 4 - 30 = 72 - 30 = 42 = M(42)$

$\therefore 4$ is a solution

$\frac{42}{6}$

One solution is $x_0 = 4$

6 mutually incongruent solutions modulo 42

$4, 4 + \left(\frac{42}{6}\right), 4 + 2\left(\frac{42}{6}\right), 4 + 3\left(\frac{42}{6}\right), 4 + 4\left(\frac{42}{6}\right), 4 + 5\left(\frac{42}{6}\right)$

$\Rightarrow 4 + 7k, \; k = 0, 1, 2, 3, 4, 5$

ie., $x \equiv \underset{k=0}{4}, \underset{k=1}{11}, \underset{k=2}{18}, 25, 32, 39 \pmod{42}$ are the solut

Euler's totient Function

Euler's totient function (also called phi (ϕ) function): counts the no. of integers less than 'n' that are co-prime to n. Thus $\phi(n)$ is the no. of $m \in Z^+$ such that $1 \le m < n$, $\gcd(m,n) = 1$

Q: Find $\phi(15)$ ($\gcd(-,15)$ from 1 to 14.)

The numbers that are less than 15 and coprime to 15 are

1 2 3 4 5 6 7 8 9 10 11 12 13 14
① ② × ④ × × ⑧ × × ⑪ × ⑬ ⑭

$= 8$

Note 1 :

Let $n = p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$ are the prime factorization of n, then $\phi(n) = n\left(1-\frac{1}{p_1}\right)\left(1-\frac{1}{p_2}\right)\cdots\left(1-\frac{1}{p_k}\right)$

Note 2 :

Euler's totient function is a multiplicative function ie., $\gcd(m,n)=1$ then $\phi(mn) = \phi(m)\cdot\phi(n)$

$\phi(36) = \phi(4\times 9) = \phi(4)\cdot\phi(9) = 2\times 6 = 12$

$4 = 2^2$, prime

$\phi(4) = 4\left(1-\frac{1}{2}\right) = 2$

$9 = 3^2$.

$\phi(9) = 9\left(1-\frac{1}{3}\right) = 9\times\frac{2}{3} = 6$

Note 3 :

If 'n' is a prime number, $\phi(n) = n-1$

Q: How many positive integers less than 900 are relatively prime to 900.

$\phi(900) = 2^2 \cdot 3^2 \cdot 5^2$

$\phi(900) = 900\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right)$

$= 240$

2	900
2	450
5	225
	45

3	900
3	300
2	100
5	50
2	10
	5

Euler's Theorem

If a and n are coprime, then $a^{\phi(n)} \equiv 1 \pmod{n}$

* a, n are coprime $\Rightarrow \gcd(a,n) = 1$
(relatively prime)

$\phi(p) = p-1$

i) Find the last 3 digit of 2^{2020}

iii) Find the remainder when 2^{50} is divided by 11.

$2^{50} \equiv ___ \pmod{11}$

Let $a=2$, $n=11$

$\gcd(2,11) = 1$

$\phi(11) = 10$

Then by eulers theorem, $2^{\phi(11)} \equiv 1 \pmod{11}$ → 2^{10}

If $a \equiv b \pmod{n}$, then $a^k \equiv b^k \pmod{n}$

$\therefore (2^{10})^5 = 1^5 \pmod{11}$

$\Rightarrow 2^{50} = 1 \bmod 11$

iv) Find remainder when $1+2+2^2+2^3+\dots+2^{100}$ is divided by 125 using Eulers theorem.

Sum: $a, ar, \dots ar^{n-1}$

$S_n = \frac{a(r^n-1)}{r-1}$, $r>1$

$1+2+2^2+2^3+\dots+2^{100} = \frac{1[2^{101}-1]}{2-1} = 2^{101}-1$

$a=1 \quad r=2$

$2^{101} - 1 \equiv ? \pmod{125}$

$\gcd(2, 125) = 1$ By F.T, $2^{\phi(125)} \equiv 1 \pmod{125}$

$\Rightarrow 2^{100} \equiv 1 \bmod 125$

$125 = 5^3$

$\phi(125) = 125\left[1 - \frac{1}{5}\right] = 100$

$\Rightarrow 2^{100} \cdot 2 \equiv 2 \pmod{125}$

$\Rightarrow 2^{101} \equiv 2 \pmod{125}$

$\Rightarrow 2^{101} - 1 \equiv 2 - 1 \pmod{125}$

$\equiv 1 \pmod{125}$

Fermat's Theorem

Let 'p' be a prime and ~~'a' be any integer such that~~ $a \in \mathbb{Z}^+$ such that $p \nmid a$ doesnot divide 'a'. $(p \nmid a)$.

Then $a^{p-1} \equiv 1 \pmod{p}$

$p \nmid a \Rightarrow \gcd(a, p) = 1 \Rightarrow a^{\phi(p)} \equiv 1 \pmod{p}$

$\Rightarrow a^{p-1} \equiv 1 \pmod{p}$

If p is a prime; then $a^p \equiv a \pmod{p}$ for any integer 'a'.

i) Use Fermat's theorem, to find $7^{222} \bmod 11$

Let $a = 7$ and $p = 11$

$11 \nmid 7 \Rightarrow$ by F.T, $7^{11-1} \equiv 1 \pmod{11}$

$\Rightarrow 7^{10} \equiv 1 \pmod{11}$

$\Rightarrow (7^{10})^{22} \equiv 1^{22} \pmod{11}$

$\Rightarrow 7^{220} \equiv 1 \pmod{11}$

$\Rightarrow 7^{220} \cdot 7^2 \equiv 1 \cdot 7^2 \pmod{11}$

$222 = 22(10) + 2$

$10 \overline{)222}$ (22, 20, 22, 20, 2)

$= 49 \pmod{11} \equiv 5 \bmod 11$

ii) Use Fermat's theorem to verify that 17 divides $11^{104}+1$

$a = 11,\ p = 17$

$17 \nmid 11 \Rightarrow$ by F.T, $11^{17-1} \equiv 1 \pmod{17}$

$\Rightarrow 11^{16} \equiv 1 \pmod{17}$

$\Rightarrow (11^{16})^6 \equiv 1 \pmod{17}$

$\Rightarrow 11^{96} \equiv 1 \pmod{17}$

$\Rightarrow 11^{96} \cdot 11^8 \equiv 1 \cdot 11^8 \pmod{17}$

$\Rightarrow 11^{104} \equiv 11^8 \pmod{17}$

$11^2 \equiv 2 \pmod{17}$

$(11^2)^2 = 11^4 \equiv 4 \pmod{17}$

$(11^4)^2 \equiv 4^2 \pmod{17}$

$11^8 \equiv 16 \pmod{17}$

$\equiv -1 \pmod{17}$

$\Rightarrow 11^{104} \equiv 11^8 \equiv -1 \pmod{17}$

$\Rightarrow 11^{104} + 1 \equiv 0 \pmod{17}$

Chinese Remainder Theorem

$n_1, n_2, \ldots, n_r \in \mathbb{Z}^+$ such that $\gcd(n_i, n_j) = 1$, for $i \neq j$.

$$\left.\begin{aligned} x &\equiv a_1 \pmod{n_1} \\ x &\equiv a_2 \pmod{n_2} \\ &\vdots \\ x &\equiv a_r \pmod{n_r} \end{aligned}\right\}$$ the system has a simultaneous soln, which is unique modulo $n_1, n_2, \ldots n_r$.

Steps to solve, CRT :-

$\gcd(n_i, N_i) = 1$

Step 1: Compute $N = n_1 \times n_2 \times \cdots \times n_k$

Step 2: For each modulus n_i, compute $N_i = \frac{N}{n_i}$

Step 3: find the multiplicative inverse M_i of N_i ; $i = 1, 2, \ldots$

ie, $M_i N_i \equiv 1 \pmod{n_i}$

Step 4: Solution is $x \equiv \sum_{i=1}^{k} a_i N_i M_i \pmod{N}$

1) Solve $x \equiv 2 \pmod 3$ Using C.R.T

$x \equiv 3 \pmod 5$

$x \equiv 2 \pmod 7$

ans: $N = 3 \times 5 \times 7$ $a_1 = 2$ $n_1 = 3$

$a_2 = 3$ $n_2 = 5$

$a_3 = 2$ $n_3 = 7$

1: $N = n_1 \times n_2 \times n_3$

$= 3 \times 5 \times 7 = 105$

2: $N_1 = \frac{N}{n_1} = \frac{105}{3} = 35$

$N_2 = \frac{N}{n_2} = \frac{105}{5} = 21$

$N_3 = \frac{N}{n_3} = \frac{105}{7} = 15$

3: To find multiplicative inverse M_i of N_i modulo n_i

i) To find inverse of $N_1 = 35$ modulo 3

$\gcd(3, 35) = 1$

By using euclidean algo,

$35 = (11)(3) + 2$

$3 = (1)(2) + 1$

$2 = (2)(1) + 0$

Extended euclidean algo,

$$1 = 3-(1)(2) = 3-(1)[35-(11)(3)]$$

$$= 3-(1)(35)+(11)(3)$$

$$=(12)(3)+(-1)(35)$$

$$1 = (-1)(35)+(12)(3)$$

$$35^{-1} \equiv (-1) \text{ modulo } 3$$

$$\equiv 2 \bmod 3 \qquad \therefore M_1 = 2.$$

ii) To find inverse of $N_2 = 21$ modulo 5

$$\gcd(5, 21) = 1$$

By using euclidean algo,

$$21 = (4)(5)+\boxed{1} \rightarrow \text{gcd}.$$

$$5 = (5)(1)+0$$

$$\begin{array}{r} 4 \\ 5\overline{)21} \\ \underline{20} \\ 1 \end{array} \qquad \begin{array}{r} 5 \\ 1\overline{)5} \\ \underline{5} \\ 0 \end{array}$$

By using extended euclidean,

$$1 = (1)(21)+(-4)(5)$$

$$21^{-1} \equiv 1 \pmod 5$$

$$\Rightarrow M_2 = 1$$

iii) To find inverse of 15 modulo.

$$15 = (2)(7)+1 \qquad \therefore M_3 = 1$$

$$7 = (7)(1)+0$$

$$1 = (1)(15)+(-2)(7)$$

4: The unique solution is given by,

$$x \equiv \sum_{i=1}^{3} a_i N_i M_i \pmod{N}$$

$$\Rightarrow x \equiv [(2)(35)(2) + (3)(21)(1) + (2)(15)(1)] \pmod{105}$$

$$\equiv 233 \pmod{105}$$

$$\equiv 23 \pmod{105}$$

$$\Rightarrow x - 23 = M(105) = 105t,\ t \in Z$$

$$\Rightarrow \boxed{x = 23 + 105t}$$

2) A certain number of objects is unknown, when divided by 3, the remainder is 1, when divided by 5, the remainder is 2, when divided by 7, the remainder is 3. Using CRT determine the no. of object.

ans.

$x_1 \equiv 1 \bmod 3$ $\quad a_1 = 1$ $\quad n_1 = 3$

$x_2 \equiv 2 \bmod 5$ $\quad a_2 = 2$ $\quad n_2 = 5$

$x_3 \equiv 3 \bmod 7$ $\quad a_3 = 3$ $\quad n_3 = 7$

70+
42
15
157

$N = 3 \times 5 \times 7 = 105$

$N_1 = \frac{105}{3} = 35$

$N_2 = \frac{105}{5} = 21$

$N_3 = \frac{105}{7} = 15$

i) To find the inverse of N_1

$N_1 = 35$ modulo 3

:

4: The unique soln,

$$x \equiv \sum_{i=1}^{3} a_i N_i M_i \pmod{N}$$

$$x \equiv [(1)(35)(2) + (2)(21)(1) + (3)(15)(1)] \bmod 105$$

$$\equiv 157 \bmod 105$$

$\equiv 52 \pmod{105}$

Ans: Solve, $x \equiv 1 \pmod 5$ $\quad a_1 = 1$ $\quad n_1 = 5$

$x \equiv 2 \pmod 6$ $\quad a_2 = 2$ $\quad n_2 = 6$

$x \equiv 3 \pmod 7$ $\quad a_3 = 3$ $\quad n_3 = 7$

$N = 5 \times 6 \times 7 = 210$